

Quantum Computing in Agriculture

GROUP-180

August 2025

Abstract: New dawn of technology and its evolution in the agricultural field have emerged as one of the greatest impacts in agriculture. But the classical approach in this field is lagging behind due to the higher demands and many other challenges. With increasing demands of technology in agriculture, there is a need to upgrade ourselves and Quantum Computing is the best alternative approach here. It has opened up new ventures for various fields, and one of the promising applications is integration of Quantum Computing in agriculture.

In quantum computing, we utilize quantum bits (qubits) that acquire properties such as superposition and entanglement. This helps in solving complex computational problems easily, effectively and at a faster rate.

In this research, we focused on applying Quadratic Unconstrained Binary Optimization (QUBO) for fertilizer optimization and Quantum Machine Learning (QML) for plant disease detection. This paper details the application of quantum computing to two key agricultural challenges: fertilizer optimization and plant disease detection.

For fertilizer optimization, the assignment problem is modeled as a Quadratic Unconstrained Binary Optimization (QUBO) problem. Constraints (e.g., one fertilizer per plot) are enforced using high-weight penalty terms. The QUBO is mapped to the Ising model for solution using Quantum Annealing (QA) or QAOA, demonstrating a scalable approach for large-scale farm management.

For plant disease detection on large datasets (e.g., PlantVillage), Quantum Machine Learning (QML) is proposed to overcome classical deep learning limitations. Features extracted from plant images are encoded into quantum states via angle encoding. Classification is performed using Quantum Support Vector Machines (QSVM) with quantum kernels or optimized Quantum Neural Networks (QNNs). This QML approach leverages superposition and entanglement to efficiently capture complex feature correlations, promising higher accuracy and faster training times for real-time diagnostics.

Abstract: Quantum Computing Applications in Sustainable Agriculture This document presents two distinct applications of quantum computing and optimization techniques to address critical challenges in sustainable agriculture: fertilizer optimization and plant/fruit disease detection.

1 Introduction

1.1 What is Quantum Computing ?

Computational demands across various research areas have been rapidly increasing recently, and thus driving the need for faster computing. Quantum computing has attracted interest since the first provable speedups in quantum algorithms and theoretical algorithms. Quantum computers can solve problems that are intractable for classical computers by leveraging the principles of quantum mechanics, specifically superposition and entanglement.

Conventional computers generally represent bits, represented as a 0 or a 1, through a series of logical gates. Applying these operations to bits allows a computer to run programs and solve problems. Physically, logical gate operations are created using transistors that require a bit or an electrical signal. Quantum computers differ fundamentally from the classical ones but share the concept of processing information using logical operations.

1.2 Quantum Computing in Agriculture

Quantum Computing technology promises to revolutionize the entire food supply chain, from the soil to the market. By tackling complex problems beyond the reach of classical computers, quantum algorithms can supercharge precision agriculture. For instance, they can process vast datasets from sensors

and drones to provide hyper-precise recommendations for water and pesticide use, reducing waste and environmental impact.

Furthermore, quantum computers can dramatically accelerate plant genetics and breeding by modeling molecular interactions and analyzing genetic data to develop resilient crop varieties faster than ever before. In the realm of logistics, these systems can optimize complex supply chains to minimize food spoilage and transport costs. Ultimately, quantum computing offers a foundational tool to address some of the most pressing challenges in global food production, paving the way for a more efficient, sustainable, and resilient agricultural future.

1.3 The Future of Fertilizer Optimization

The global food system is at a crossroads, facing the immense task of feeding a growing population without destroying our planet. For generations, we have relied on the heavy use of fertilizers to increase crop yields. But this solution has come at a steep price, leading to polluted waterways, damaged soil, and increased greenhouse gas emissions. The way we currently manage fertilizers is simply inefficient, creating a major obstacle to building a truly sustainable food system. We urgently need a smarter and more precise way to farm.

This is where the power of quantum computing comes in. It is not just a new technology; it is a tool that can help us solve one of our most pressing human problems. Algorithms like the Quantum Approximate Optimization Algorithm (QAOA) can tackle incredibly complex challenges that are impossible for today's computers. By using these tools, we can move away from wasteful, one-size-fits-all farming and toward a future where every single plant gets exactly what it needs, when it needs it. This shift isn't just about efficiency; it is about making our food system more responsible, resilient, and in harmony with the environment.

2 Literature Review

The push to use advanced computing in agriculture is a hot topic in recent trends and for good reason. It is all about finding a smarter way to grow our food. While regular computers have done a lot for farming, they just cannot handle the super-complex problems that come with nature. That is why a lot of researchers are turning to quantum computing to do the job.

A big part of the work there is focused on using quantum algorithms for tough optimization problems. For example, experts like Farhi and Hadfield have been pioneering a method called the Quantum Approximate Optimization Algorithm (QAOA), showing how it could find solutions to real-world problems. Although this started out as just theory, a lot of new papers are coming out that are trying to apply these ideas directly to things like how we manage resources on a farm.

Scientists are also using quantum computers to get a closer look at the smallest parts of agriculture. Think about what happens at the molecular level with soil and plants. We can now use quantum computers to run simulations that were impossible before. This research is giving us a whole new understanding of things such as how nitrogen is fixed in the soil, which could help us to invent more effective and eco-friendly fertilizers.

And it is not just about what happens in the field. The literature is showing that quantum computing could change the entire food chain, from planting to getting food on our tables. Researchers are talking about how these computers could be used to create better weather forecasts, develop stronger crop varieties, and even make supply chains run more smoothly. While we are still in the early stages, the research so far paints a pretty compelling picture: quantum computing is not just a gimmick; it is a foundational tool that could help us create a more efficient and sustainable future for farming.

3 Methodology

3.1 Problem Identification

Modern agriculture faces a critical challenge in the efficient use of fertilizers. Excessive application of fertilizers increases crop yield only marginally but leads to serious consequences such as soil degradation, groundwater pollution, and higher input costs. On the other hand, insufficient application fails to meet the nutrient requirements of crops, resulting in reduced productivity. Therefore, an optimization framework is required to determine the *right type*, *right amount*, and *right timing* of fertilizer application.

Mathematically, the problem can be defined as finding an allocation vector

$$x = (x_1, x_2, \dots, x_n), \quad x_i \in \{0, 1\}$$

where each x_i represents the decision to apply fertilizer i (at a chosen level). The goal is to minimize the total cost while meeting crop nutrient demand D .

Constraints include:

- Soil nutrient status S (e.g., nitrogen, phosphorus, potassium levels).
- Crop nutrient requirement D for optimal growth.
- Fertilizer nutrient composition N_i and cost c_i .
- Environmental limits (e.g., nitrogen runoff, carbon footprint).

At this stage, only the basic cost-demand balance is considered:

$$\text{Minimize } C(x) = \sum_{i=1}^n c_i x_i \quad \text{subject to} \quad \sum_{i=1}^n N_i x_i \geq D$$

The extended version of the problem is intended to incorporate multiple nutrients and non-linear environmental penalties, but this is still under development.

3.2 Required Data

To build our fertilizer optimization model, we need clear and organized data from several key areas. The main types of data required are:

- **Soil Properties:** The starting amount of nutrients in the soil (such as Nitrogen, Phosphorus, and Potassium), along with the soil's acidity (pH), how much water it holds, and its organic content.
- **Crop Needs:** The specific amount of each nutrient a particular type of crop requires to grow, shown as its demand (D) for Nitrogen, Phosphorus, and Potassium.
- **Fertilizer Details:** The nutrient makeup (N_i) of each kind of fertilizer, any rules for how it can be used, and its price (c_i).
- **Environmental Limits:** Rules to prevent over-fertilization, such as the maximum amount of nutrients allowed to seep into groundwater or wash away, and the penalties for breaking these rules.

For this initial study, we use a simple setup with a small number of fertilizer options (3 to 5). Each one has a set price and known nutrient value. While real data from farm records or government reports could be used in the future for better accuracy, this paper uses example data to first test if the quantum optimization method works.

3.3 Mathematical model

The fertilizer allocation problem is modeled as an optimization task with both economic and environmental objectives. The main aim is to determine the right combination of fertilizers that satisfies crop nutrient requirements while minimizing overall cost and over-application penalties.

Let the decision variables be defined as:

$$x_i \in \{0, 1\}, \quad i = 1, 2, \dots, n$$

where $x_i = 1$ indicates that fertilizer i is applied, and $x_i = 0$ otherwise. Each fertilizer has an associated cost c_i and a nutrient contribution N_i .

The objective function can be expressed as:

$$\min F(x) = \sum_{i=1}^n c_i x_i + \lambda \left(D - \sum_{i=1}^n N_i x_i \right)^2$$

where D denotes the crop nutrient demand and λ is a penalty factor to handle under- or over-supply.

This formulation naturally leads to a Quadratic Unconstrained Binary Optimization (QUBO) model, which can be mapped onto a quantum Hamiltonian:

$$H = \sum_i a_i Z_i + \sum_{i < j} b_{ij} Z_i Z_j$$

where Z_i represents the Pauli-Z operator acting on qubit i , and the coefficients a_i , b_{ij} encode costs and nutrient constraints.

The quantum algorithm chosen for this work is the Quantum Approximate Optimization Algorithm (QAOA). The variational form of QAOA alternates between applying the problem Hamiltonian H and a mixing Hamiltonian $H_m = \sum_i X_i$, where X_i is the Pauli-X operator. The state is then measured, and classical optimization is used to refine the variational parameters.

At this stage, only small-scale simulations are considered (e.g., 3–5 fertilizers), and the formulation is still being extended to include additional constraints such as environmental regulations and multi-nutrient balance.

4 What is a QUBO Problem?

QUBO stands for **Quadratic Unconstrained Binary Optimization**. It is a mathematical model used to represent many difficult optimization problems in a unified way.

The idea is to find binary variables $x_i \in \{0, 1\}$ that minimize a quadratic objective function. The general form is:

$$\min_{x \in \{0, 1\}^n} H(x) = \sum_{i < j} Q_{ij} x_i x_j + \sum_i Q_{ii} x_i$$

Here:

- $x = (x_1, x_2, \dots, x_n)$ is a vector of binary decision variables.
- Q is a coefficient matrix that encodes the problem.
- The quadratic terms $\sum_{i < j} Q_{ij} x_i x_j$ represent interactions between pairs of variables.
- The linear terms $\sum_i Q_{ii} x_i$ represent individual contributions (biases) of each variable.

4.1 How does it work?

The power of QUBO lies in its flexibility: many optimization problems (maximization, minimization, with constraints) can be reformulated into this single quadratic structure. Both the **objective function** and **constraints** are expressed through the coefficients of the matrix Q .

4.2 Connection to Quantum Computing

QUBO is closely related to the **Ising model** from physics, which describes the energy of interacting spins (magnetic particles). Quantum annealers, such as those built by D-Wave, are designed to find the lowest-energy (ground) state of an Ising system. This means that by mapping a QUBO problem to an Ising model, a quantum annealer can efficiently search for the best (or near-best) solution.

5 Problem 1: Fertilizer Optimization as a QUBO Problem

Now we show how QUBO can be applied to a farming scenario.

5.1 Problem Statement

A farmer has three plots of land (P_1, P_2, P_3) and two types of fertilizer (F_1, F_2). The goal is to decide which fertilizer to apply to each plot to maximize the overall profit.

5.2 Decision Variables

We define binary variables to represent our decisions:

$$\begin{aligned} x_1 &= 1 \text{ if } P_1 \text{ gets } F_1, 0 \text{ otherwise,} \\ x_2 &= 1 \text{ if } P_1 \text{ gets } F_2, 0 \text{ otherwise,} \\ x_3 &= 1 \text{ if } P_2 \text{ gets } F_1, 0 \text{ otherwise,} \\ x_4 &= 1 \text{ if } P_2 \text{ gets } F_2, 0 \text{ otherwise,} \\ x_5 &= 1 \text{ if } P_3 \text{ gets } F_1, 0 \text{ otherwise,} \\ x_6 &= 1 \text{ if } P_3 \text{ gets } F_2, 0 \text{ otherwise.} \end{aligned}$$

5.3 Objective Function (Profit Maximization)

Profit = Total Revenue – Total Cost. Since QUBO is a minimization model, we minimize the negative profit.

Revenues. Suppose expected revenues are:

$$R_{11}, R_{12}, R_{21}, R_{22}, R_{31}, R_{32}$$

corresponding to each plot-fertilizer combination.

$$\text{Total Revenue} = R_{11}x_1 + R_{12}x_2 + R_{21}x_3 + R_{22}x_4 + R_{31}x_5 + R_{32}x_6$$

Costs. Fertilizer costs are:

$$C_1 \text{ for } F_1, \quad C_2 \text{ for } F_2.$$

$$\text{Total Cost} = C_1(x_1 + x_3 + x_5) + C_2(x_2 + x_4 + x_6)$$

Penalty. If F_1 is used on P_1 and F_2 is used on P_2 simultaneously, an extra cost P_{12} is incurred:

$$\text{Penalty} = P_{12}x_1x_4$$

Final Objective.

$$H(x) = -(R_{11}x_1 + \dots + R_{32}x_6) + C_1(x_1 + x_3 + x_5) + C_2(x_2 + x_4 + x_6) + P_{12}x_1x_4$$

5.4 Constraints (One Fertilizer per Plot)

Each plot must receive exactly one fertilizer. For plot P_1 :

$$x_1 + x_2 = 1$$

This constraint can be added as a penalty term:

$$A(x_1 + x_2 - 1)^2$$

where A is a large constant. Similarly:

$$A(x_3 + x_4 - 1)^2, \quad A(x_5 + x_6 - 1)^2$$

5.5 Numerical Example and QUBO Formulation

Suppose:

$$R_{11} = 10, R_{12} = 8, R_{21} = 12, R_{22} = 11, R_{31} = 9, R_{32} = 10$$

$$C_1 = 3, C_2 = 2, P_{12} = 5, A = 20$$

Linear coefficients:

$$Q_{11} = -27, Q_{22} = -26, Q_{33} = -29, Q_{44} = -29, Q_{55} = -26, Q_{66} = -28$$

Quadratic coefficients:

$$Q_{12} = 40, \quad Q_{34} = 40, \quad Q_{56} = 40, \quad Q_{14} = 5$$

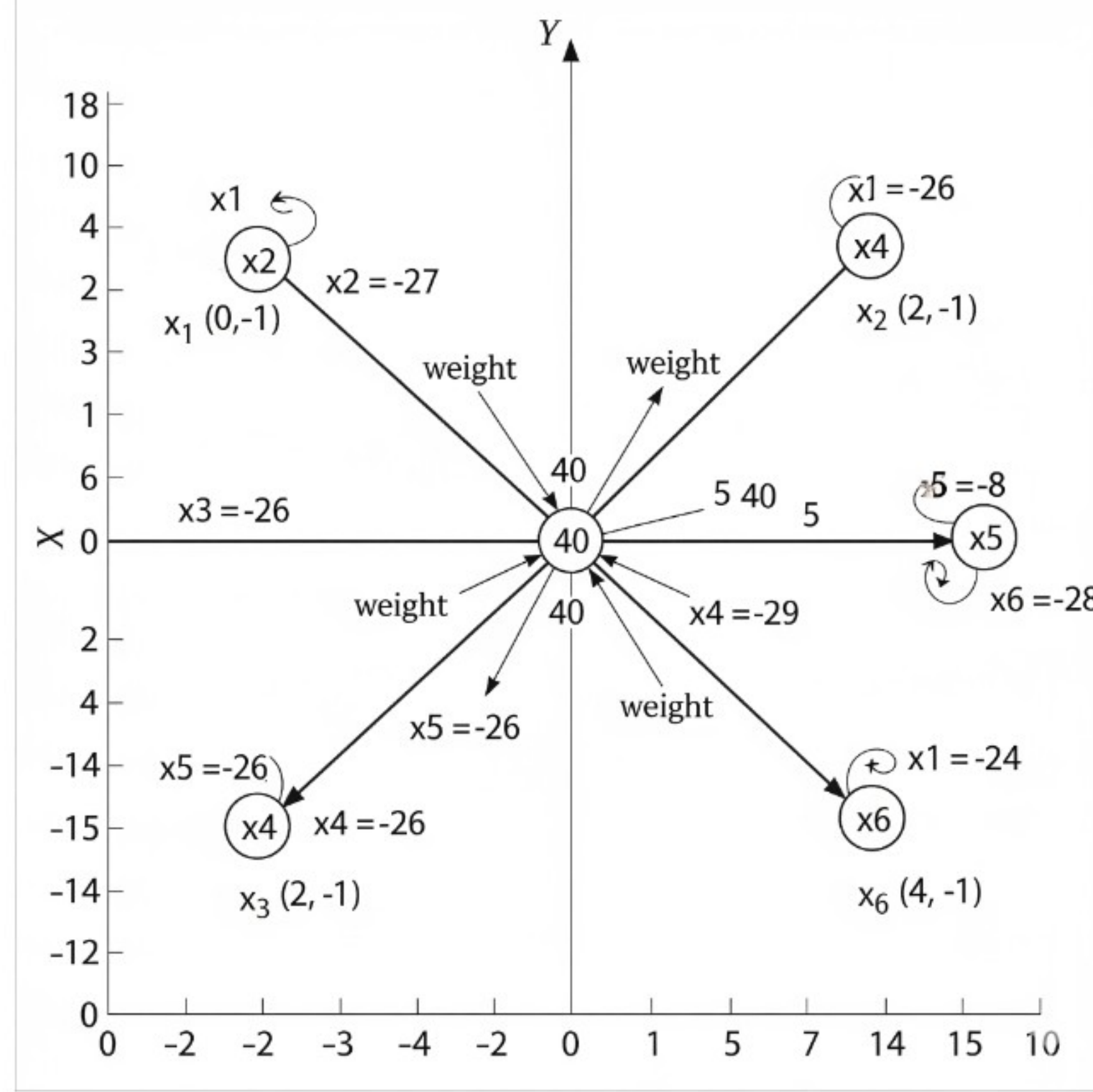


Figure 1: QUBO Coupling Graph plotted on a 2D plane, using the coordinates and weights derived from the above numerical example.

5.6 Graph Interpretation for QUBO Fertilizer Optimization

The graph visualizes the entire QUBO objective function $H(\mathbf{x}) = \mathbf{x}^T \mathbf{Q} \mathbf{x}$:

- **Vertical Constraint Edges (Weight $Q_{ij} = 40$):**
 - These are the high-weight connections between (x_1, x_2) , (x_3, x_4) , and (x_5, x_6) .
 - They represent a strong ferromagnetic coupling and enforce the crucial **one-fertilizer-per-plot constraint** (e.g., $x_1 + x_2 = 1$). The large positive weight ensures the system strongly penalizes solutions where two fertilizers are chosen for one plot ($x_i = 1, x_j = 1$) or zero are chosen ($x_i = 0, x_j = 0$), forcing the solution path into the feasible region.
- **Diagonal Penalty Edge (Weight $Q_{14} = 5$):**
 - This is the connection between x_1 and x_4 .
 - It represents the **extra cost incurred** (Penalty = $P_{12}x_1x_4$) if Fertilizer 1 is chosen for Plot 1 ($x_1 = 1$) and Fertilizer 2 is chosen for Plot 2 ($x_4 = 1$) simultaneously.
- **Node Weights (Self-Loops Q_{ii}):**
 - The negative weights on the nodes (e.g., $x_3 = -29$) act as **local magnetic fields**.
 - These fields drive the system toward states where variables with larger *negative* weights are set to 1. Since the overall objective is to **minimize** $H(\mathbf{x})$, activating a variable with a large negative Q_{ii} coefficient helps minimize $H(\mathbf{x})$ and thus maximizes the overall profit.

5.7 Solution

For a small problem with only 6 binary variables, a classical solver can find the solution by brute force. But for larger farms (many plots and fertilizer types), the number of variables grows rapidly, making it a strong candidate for a quantum annealer.

The optimal binary vector $x = (x_1, \dots, x_6)$ will give the best assignment of fertilizers to plots.

For example, one feasible solution is:

$$x_2 = 1, x_3 = 1, x_6 = 1$$

which means:

$$P_1 \rightarrow F_2, \quad P_2 \rightarrow F_1, \quad P_3 \rightarrow F_2$$

with high overall profit.

6 QUBO Algorithm

6.1 Problem definition

A Quadratic Unconstrained Binary Optimization (QUBO) problem is

$$\min_{x \in \{0,1\}^n} f(x) = x^\top Q x = \sum_{i=1}^n \sum_{j=1}^n Q_{ij} x_i x_j, \quad (1)$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric (diagonal entries capture linear terms). If the problem has constraints, add quadratic penalty terms so that the objective remains unconstrained:

$$\tilde{Q} = Q + \lambda Q_{\text{penalty}}, \quad \lambda \gg 0 \text{ (a suitably large penalty).}$$

6.2 Mapping to Ising form

Quantum annealers and many algorithms use spin variables $s_i \in \{-1, 1\}$. Convert via

$$x_i = \frac{1 + s_i}{2} \quad \Leftrightarrow \quad s_i = 2x_i - 1.$$

Under this transformation the Hamiltonian (energy) becomes an Ising model

$$H(s) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \text{constant},$$

with coefficients h_i, J_{ij} obtained by algebraic expansion of $x^\top Q x$.

6.3 High-level algorithm (steps)

Input: QUBO matrix Q and (optional) constraints, solver choice (QA or QAOA), solver parameters
Output: Approximate minimizer(s) x^* and objective values
begin
 // 1. Constraint handling
 If problem has constraints, build penalty matrix Q_{penalty} and set $\tilde{Q} \leftarrow Q + \lambda Q_{\text{penalty}}$;
 // 2. Preprocessing (optional)
 Apply reductions: fix variables with obvious value, scale entries, remove near-zero couplings;
 // 3. Choose solver
 if $\text{solver} == \text{QA}$ (*quantum annealer*) **then**
 Convert $\tilde{Q}$ to Ising coefficients h, J ;
 Find a minor-embedding of the Ising graph into hardware graph (chain construction);
 Tune anneal time, number of reads, and chain strengths;
 Run annealing: collect many samples and energies;
 else if $\text{solver} == \text{QAOA}$ (*gate-based*) **then**
 Convert $\tilde{Q}$ into cost Hamiltonian H_C (Pauli-Z form);
 Choose QAOA depth p and initial parameters γ, β ;
 for *classical optimization steps until convergence* **do**
 Prepare variational circuit $U(\gamma, \beta)$ and measure samples;
 Estimate expected energy $\langle H_C \rangle$ and update parameters via classical optimizer;
 end
 Sample optimized circuit repeatedly to obtain bitstrings;
 // 4. Postprocessing
 Decode/minority-vote chain breaks (for QA) and map spins back to bits x ;
 Optionally apply classical local improvement (e.g., greedy bit flips) to each sample;
 Return best-found solution(s) and statistics (best energy, frequency, distribution).
end

Algorithm 1: General QUBO-to-Quantum solver workflow

6.4 Pseudocode (compact)

Input: Q , solver parameters
Output: Best solution x^*
Preprocess Q ;
Add penalties if needed: $\tilde{Q} \leftarrow Q + \lambda Q_{\text{penalty}}$;
Convert to Ising: compute h, J ;
if *Quantum Annealing* **then**
 Embed graph, set chain strengths, run many anneals, collect samples;
end
else if *QAOA* **then**
 Build cost Hamiltonian H_C , choose depth p , optimize γ, β , sample final circuit;
end
Classical postprocess (decode, repair, local search);
Return best sample x^* and energy $f(x^*)$;

Algorithm 2: Compact QUBO solver pseudocode

6.5 Notes on practical implementation

- **Penalty tuning:** Use progressively larger λ if constraint violations are observed. Too large λ can make optimization ill-conditioned.
- **Embedding:** Minor-embedding is NP-hard; practical embedding heuristics (e.g., Cai et al.'s heuristic) are used. Chains must have appropriate strengths to avoid broken chains.
- **Sampling budget:** Use many reads/samples to estimate rugged landscapes. Combine quantum samples with classical improvement (hybrid strategies).

- **QAOA depth:** Increasing p generally improves expressivity but increases circuit depth and classical optimization difficulty.

6.6 Mapping example: QUBO to Ising (explicit)

For one-to-one mapping, substitute $x_i = (1 + s_i)/2$ into $x^\top Qx$:

$$x^\top Qx = \frac{1}{4} \sum_{i,j} Q_{ij} (1 + s_i)(1 + s_j) = \underbrace{\frac{1}{4} \sum_{i,j} Q_{ij} \text{const}} + \frac{1}{2} \sum_i \left(\sum_j Q_{ij} \right) s_i + \frac{1}{4} \sum_{i \neq j} Q_{ij} s_i s_j.$$

Collect terms into linear biases h_i and couplers J_{ij} for the Ising Hamiltonian

$$H(s) = \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j + \text{const},$$

with $h_i = \frac{1}{2} \sum_j Q_{ij}$ (plus diagonal contributions handled consistently) and $J_{ij} = \frac{1}{4}(Q_{ij} + Q_{ji})$.

6.7 Complexity and guarantees

The QUBO problem is NP-hard in general. Quantum methods (QA, QAOA) provide heuristic or probabilistic procedures that can offer improved sampling of low-energy states for certain problem families, but do not guarantee polynomial-time optimal solutions for arbitrary QUBOs.

6.8 References and further reading

(Include references in your paper to QAOA, quantum annealing, and embedding heuristics, e.g., Farhi et al. (2014) for QAOA and D-Wave technical papers for annealing and embedding.)

7 Problem 2: Plant and Fruit Disease Detection using Quantum Computing

7.1 Problem Identification

Agriculture forms the backbone of the global economy, especially in developing countries like India, where more than 60% of the population depends on farming for livelihood. Plant and fruit diseases are among the primary causes of crop loss, leading to reduced yield, poor quality, and economic instability for farmers. According to the Food and Agriculture Organization (FAO), around 20–40% of global crop production is lost annually due to pests and diseases.

Traditional disease detection relies on expert visual inspection, which is time-consuming, subjective, and impractical for large-scale farming. Classical machine learning and deep learning methods such as Convolutional Neural Networks (CNNs) have shown promising results in automatic plant disease detection using image datasets. However, with increasing dataset sizes and higher feature complexities, classical algorithms face challenges such as:

- High computational cost during training.
- Poor scalability to large datasets.
- Difficulty in capturing highly correlated features (color + texture + shape).

Quantum computing provides a potential breakthrough by utilizing superposition and entanglement to represent and process high-dimensional data efficiently. The integration of Quantum Machine Learning (QML) with agricultural disease detection could accelerate training and improve classification accuracy.

7.2 Data Requirement

For this research, the PlantVillage dataset is considered, which consists of more than 50,000 leaf images across 38 classes of healthy and diseased plants. For fruit crops, classes include healthy and infected leaves of apple, grape, mango, banana, and tomato. The raw dataset requires preprocessing before being used in quantum models:

1. **Image Preprocessing:** Resize images to a fixed dimension (e.g., 128×128), convert to grayscale or normalized RGB.
2. **Noise Removal:** Apply Gaussian filtering or histogram equalization to improve quality.
3. **Feature Extraction:** Extract color histograms, texture descriptors (GLCM), and shape features of infected regions.
4. **Dimensionality Reduction:** Apply Principal Component Analysis (PCA) to reduce feature space to a lower dimension suitable for quantum encoding.

After preprocessing, each image is represented by a feature vector:

$$\mathbf{x} = [x_1, x_2, x_3, x_4, \dots, x_d]$$

where d is the reduced dimension, typically $4 \leq d \leq 8$ for quantum encoding.

7.3 Mathematical Model

The extracted features are mapped into the quantum Hilbert space. Each feature x_i is encoded into a qubit state using angle encoding:

$$|\psi_i\rangle = \cos(x_i)|0\rangle + \sin(x_i)|1\rangle$$

For d features, the joint quantum state is obtained through the tensor product:

$$|\Psi\rangle = \bigotimes_{i=1}^d |\psi_i\rangle$$

Thus, the feature vector $\mathbf{x}$ is represented in a 2^d dimensional Hilbert space, allowing the quantum model to explore correlations between features more effectively.

7.4 Methodology

The proposed methodology for disease detection follows the steps below:

1. **Classical Preprocessing:** Clean and normalize leaf images to extract key features.
2. **Quantum Encoding:** Map extracted features into qubits using angle encoding.
3. **Model Selection:** Apply Quantum Support Vector Machine (QSVM) and Quantum Neural Network (QNN) approaches.
4. **Training and Optimization:** Use quantum kernel estimation and parameterized quantum circuits to optimize classification.
5. **Measurement and Classification:** Measure qubit states to assign a disease label (e.g., healthy, blight, rust, mildew).

Quantum Support Vector Machine (QSVM)

The kernel trick in classical SVM is replaced by quantum kernels:

$$K(x_i, x_j) = |\langle \psi(x_i) | \psi(x_j) \rangle|^2$$

The decision function is expressed as:

$$f(x) = \text{sign} \left(\sum_{i=1}^N \alpha_i y_i K(x_i, x) + b \right)$$

This allows for separation of complex and overlapping feature distributions, which is common in diseased vs healthy plant leaves.

Quantum Neural Network (QNN)

A QNN uses parameterized quantum circuits (PQCs) with rotation gates and entanglement layers:

$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

Training involves minimizing a cost function:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

where $\hat{y}_i$ is the predicted probability of disease presence. Quantum backpropagation (using parameter-shift rule) is applied for optimization.

7.5 Illustrating step-by-step process to detect a disease in a fruit leaf Using QSVM Model:



Figure 2: **Image Preprocessing:** This image shows the initial steps of taking a raw image of a diseased apple leaf and processing it by resizing and converting it to grayscale.

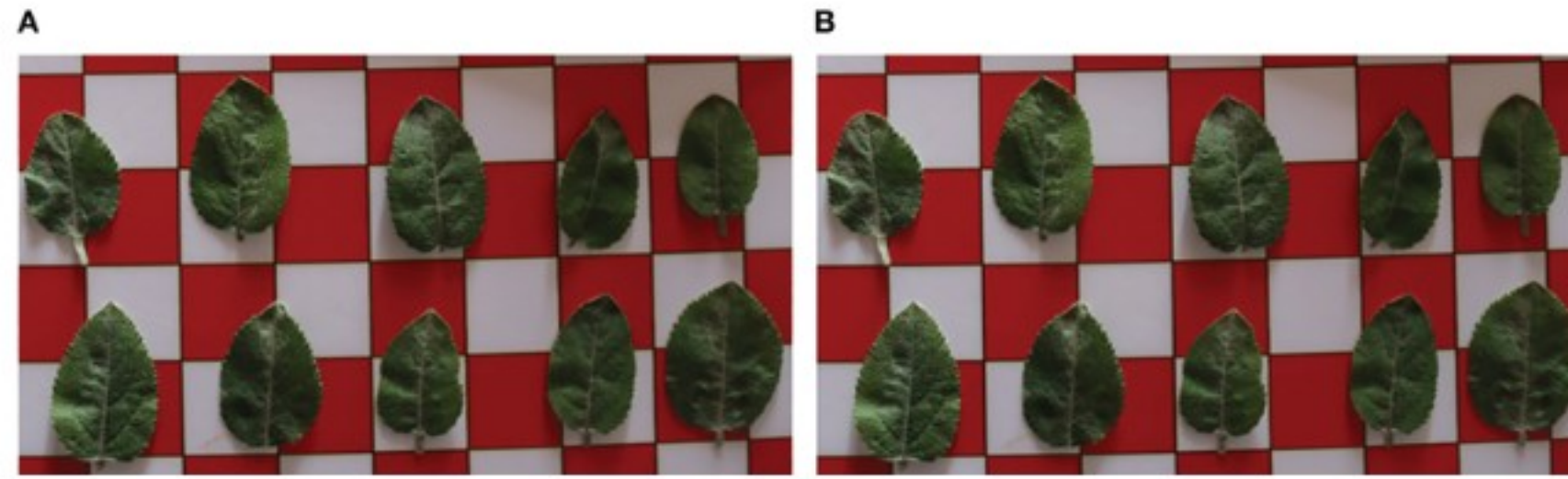


Figure 3: **Noise Removal:** Here is an example of a Gaussian filter being applied to a leaf image to remove unwanted noise and improve image quality for better feature extraction.

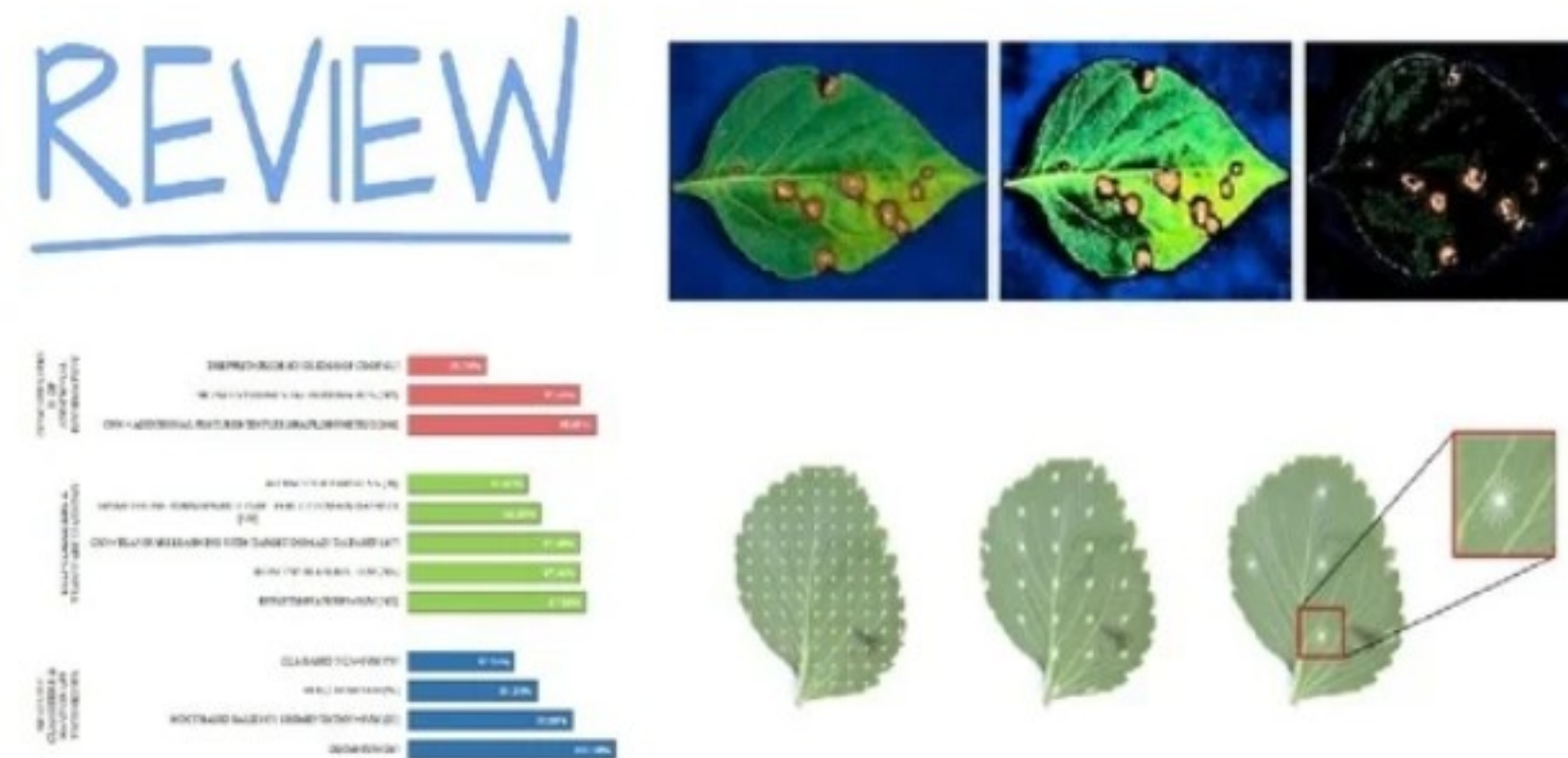


Figure 4: **Feature Extraction:** This image demonstrates how key features like color histograms and texture descriptors (GLCM) are extracted from the processed leaf image.

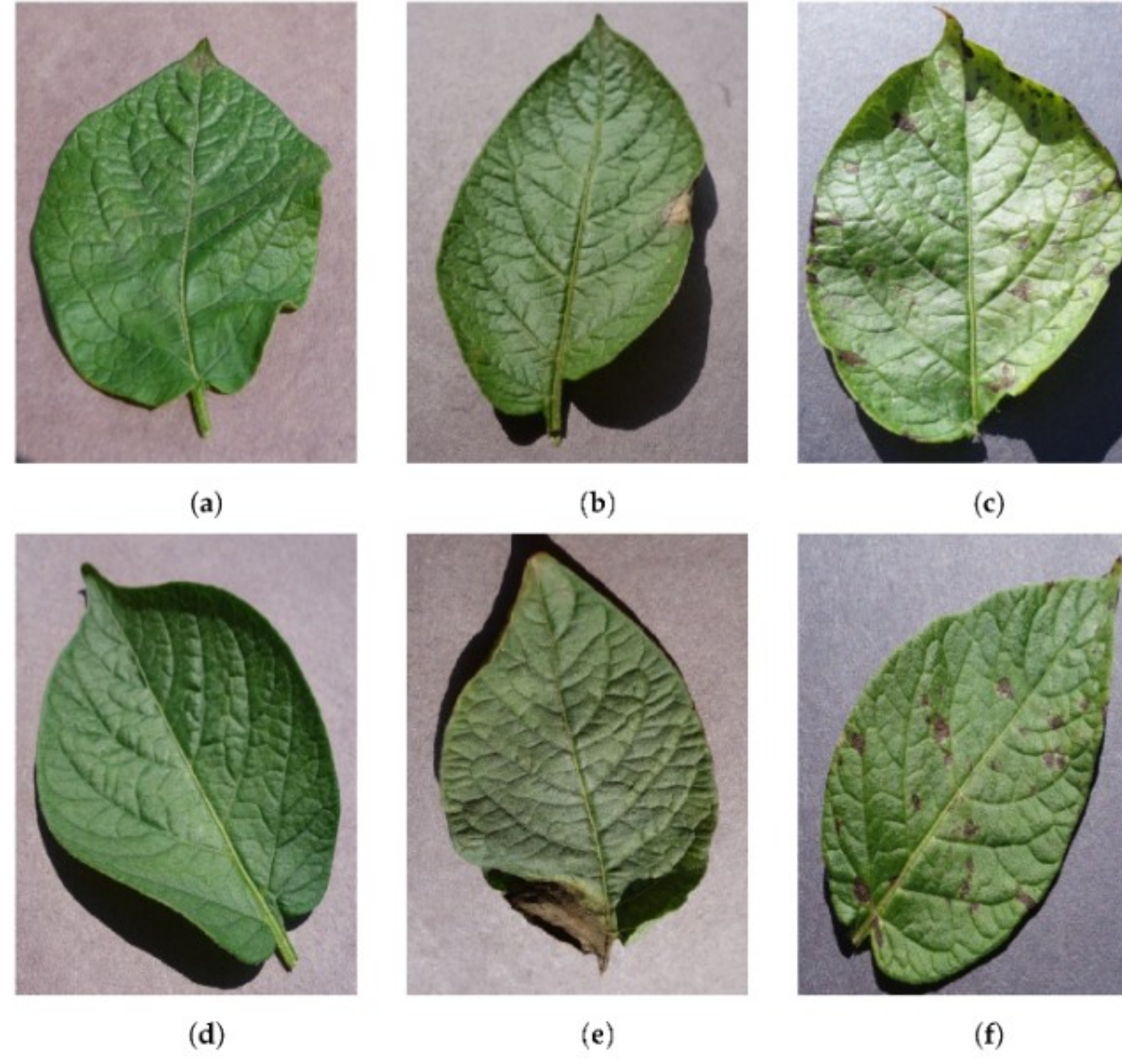


Figure 5: **Dimensionality Reduction (PCA)**: This visual represents how Principal Component Analysis (PCA) reduces the high-dimensional feature vector into a smaller, more manageable dimension suitable for quantum algorithms.

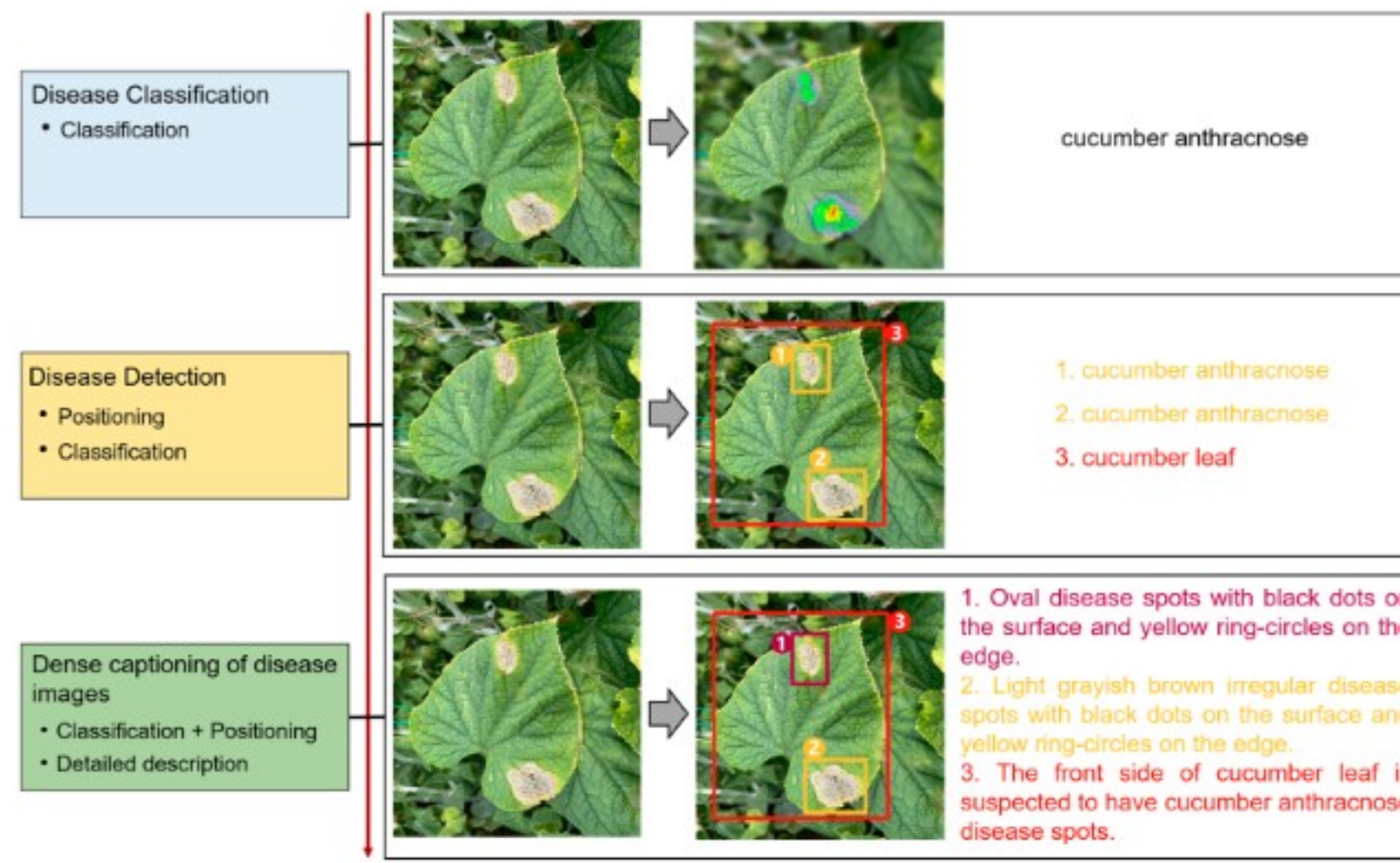


Figure 6: **Quantum Encoding**: This image shows the process of mapping the classical feature vector into a quantum state using angle encoding.

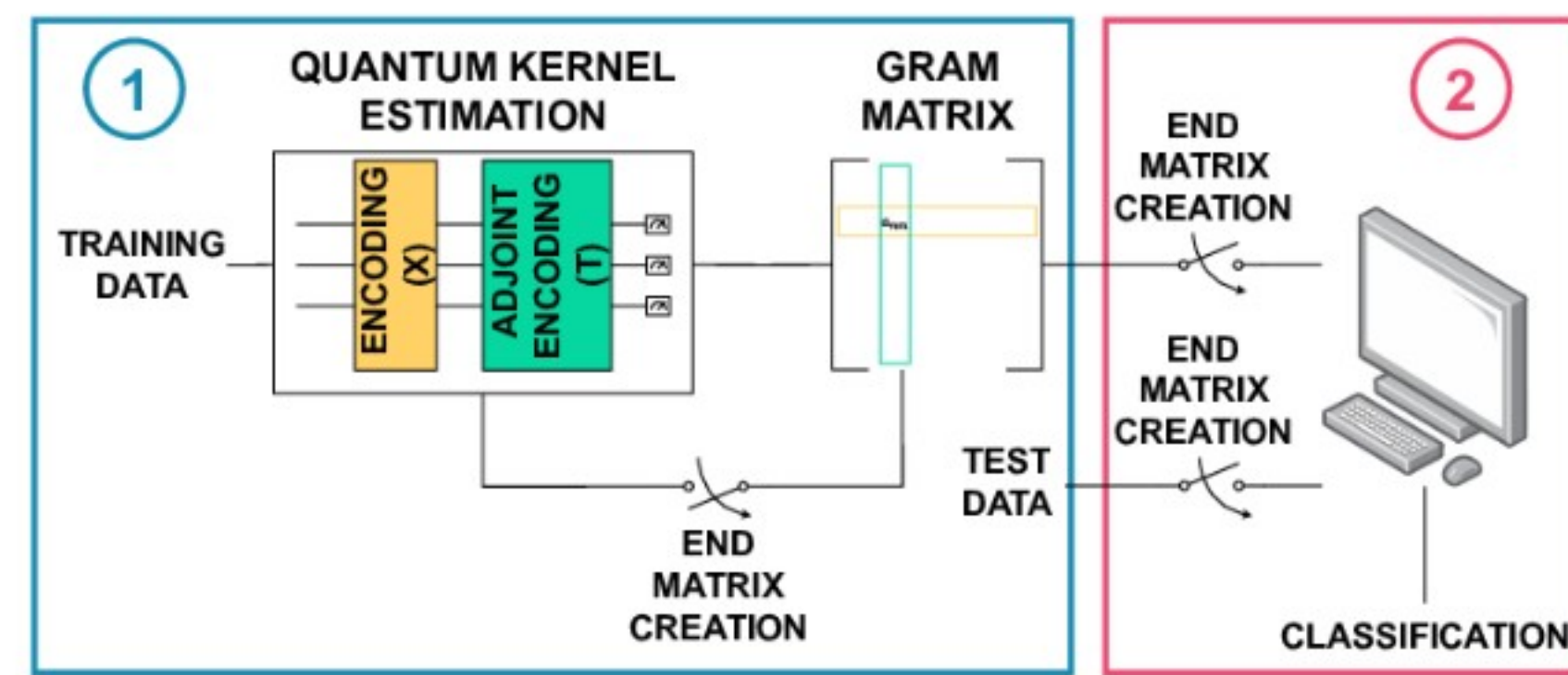


Figure 7: **Quantum Model Selection (QSVM)**: Here is visual representation of the quantum machine learning model, QSVM, which is used for classification.

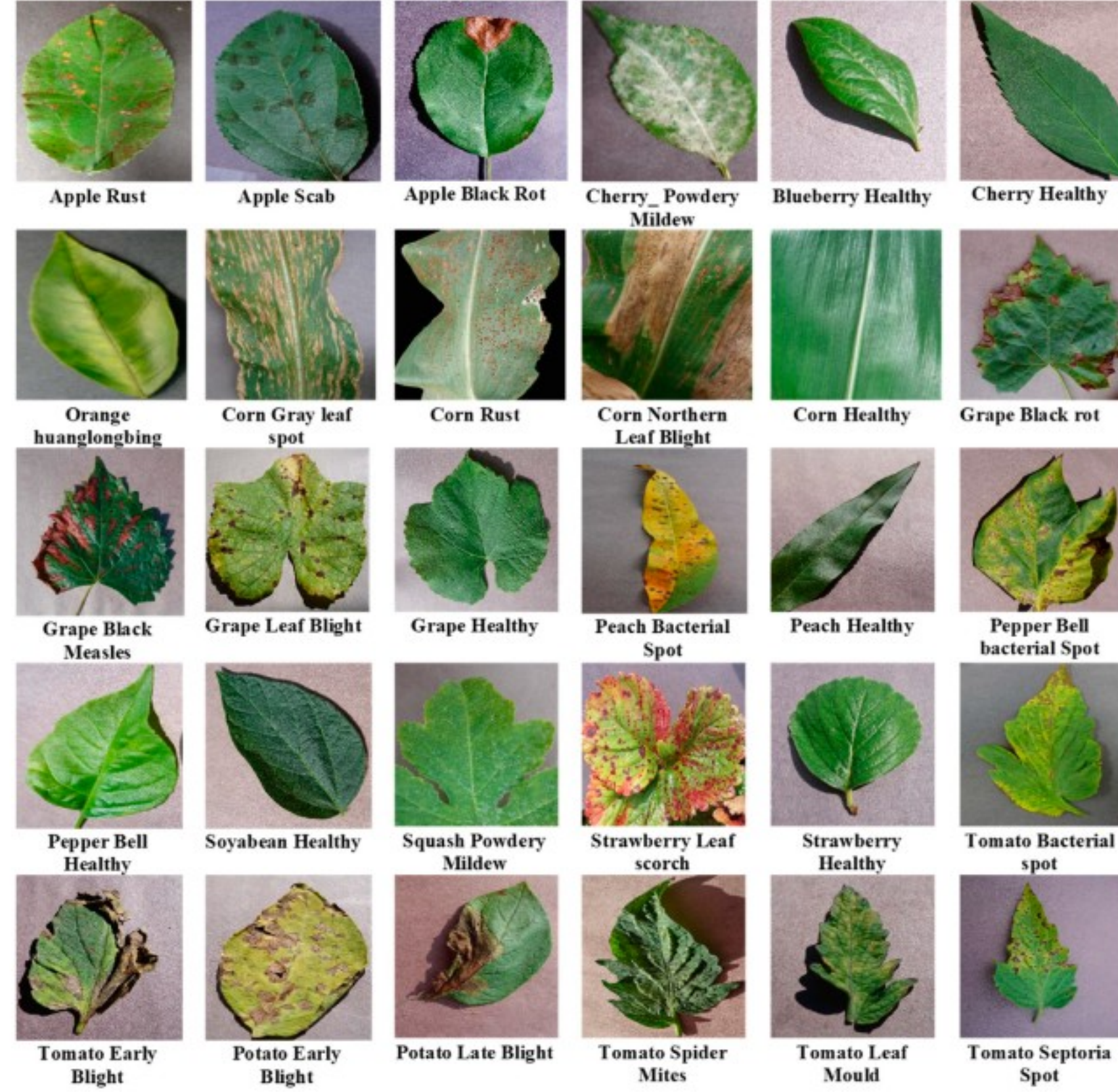


Figure 8: **Measurement and Classification:** This final image illustrates how the quantum state is measured to determine the final classification, assigning a label like "healthy" or "diseased."

7.6 How Mathematics is Used

- **Linear Algebra:** Representation of qubits and operations as vectors and matrices.
- **Tensor Products:** Used to represent multi-qubit systems and feature interactions.
- **Trigonometry:** Encoding of classical data into quantum states via sine and cosine rotations.
- **Kernel Methods:** Quantum kernels based on inner products improve non-linear classification.
- **Optimization:** Loss functions and gradient descent techniques are applied in QNNs.

7.7 Evaluation Metrics

The performance of the model is evaluated using:

- Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- F1-score = $2 \times \frac{Precision \times Recall}{Precision + Recall}$

Where TP, TN, FP, FN represent true positives, true negatives, false positives, and false negatives respectively.

7.8 Quantum Advantage

Quantum computing provides significant benefits over classical approaches in disease detection:

1. **Superposition:** Encodes multiple disease features simultaneously.
2. **Entanglement:** Captures interdependence between features (e.g., color + texture).
3. **Efficient Kernels:** Quantum kernels separate classes in high-dimensional feature space.
4. **Scalability:** Handles large datasets with fewer computational resources.
5. **Speed:** Training time is reduced compared to classical deep learning for certain tasks.

This makes quantum-enhanced classification models a strong candidate for real-time disease detection in agriculture.

7.9 Conclusion

The integration of quantum computing into plant and fruit disease detection provides a powerful alternative to traditional machine learning. By leveraging quantum encoding, QSVM, and QNN, the proposed methodology can detect diseases with higher accuracy and efficiency. Future work will focus on hybrid quantum-classical models and testing on real-world agricultural datasets.

8 Justification for Quantum Support Vector Machine (QSVM)

The Quantum Support Vector Machine (QSVM) was selected as the core classification model for plant disease detection due to its superior efficiency and stability in high-dimensional feature spaces, offering key advantages over both classical and other quantum approaches:

- **Quantum Feature Advantage:** QSVM utilizes **Quantum Feature Maps** to implicitly project input data into an exponentially large Hilbert space (2^d). This allows it to capture complex, non-linear correlations in the image feature data (like GLCM and color histograms) that are often inaccessible to classical kernels, leading to higher classification accuracy for fine-grained disease detection.
- **Superior Stability over QNNs:** Unlike Quantum Neural Networks (QNNs), the QSVM completely avoids the critical **Barren Plateau problem**. Since the QSVM’s training primarily involves the classical optimization of support vector weights, the hybrid training process is far more stable, faster, and reliable than the gradient-based optimization required for QNNs.
- **Computational Efficiency vs. CNNs:** While Classical Convolutional Neural Networks (CNNs) achieve high performance, they demand extensive computational resources (GPUs) and massive datasets to train millions of parameters. QSVM offers a path toward quantum speedup, leveraging a fixed, small quantum circuit structure for kernel calculation, making it significantly more energy-efficient and scalable than training deep classical models.
- **Reduced Data Dependency:** The kernel-based approach of QSVM, rooted in statistical learning theory, often achieves competitive performance with significantly smaller training datasets compared to data-hungry deep learning models (CNNs). This is a critical advantage when labeled agricultural imagery is scarce or expensive to acquire.
- **Clear Theoretical Foundation:** QSVM is based on the rigorous Vapnik–Chervonenkis (VC) theory, providing a clear, theoretically grounded boundary for generalization performance. This makes the QSVM model more predictable and less dependent on heuristic fine-tuning compared to the complex, less interpretable architectures of traditional deep neural networks.

9 Quantum Algorithm and Flowchart

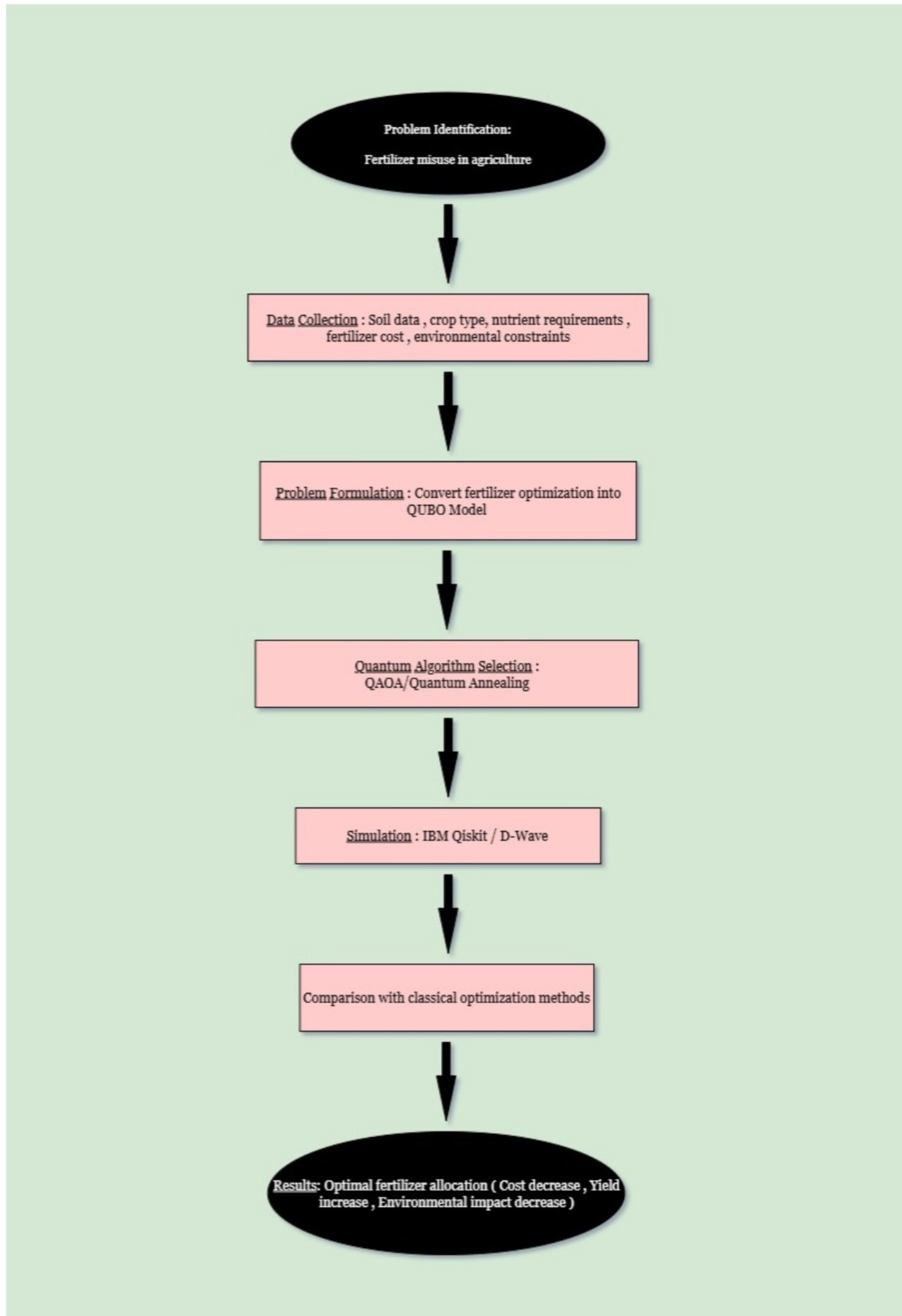


Figure 9: An illustration of the quantum-powered approach to fertilizer optimization.

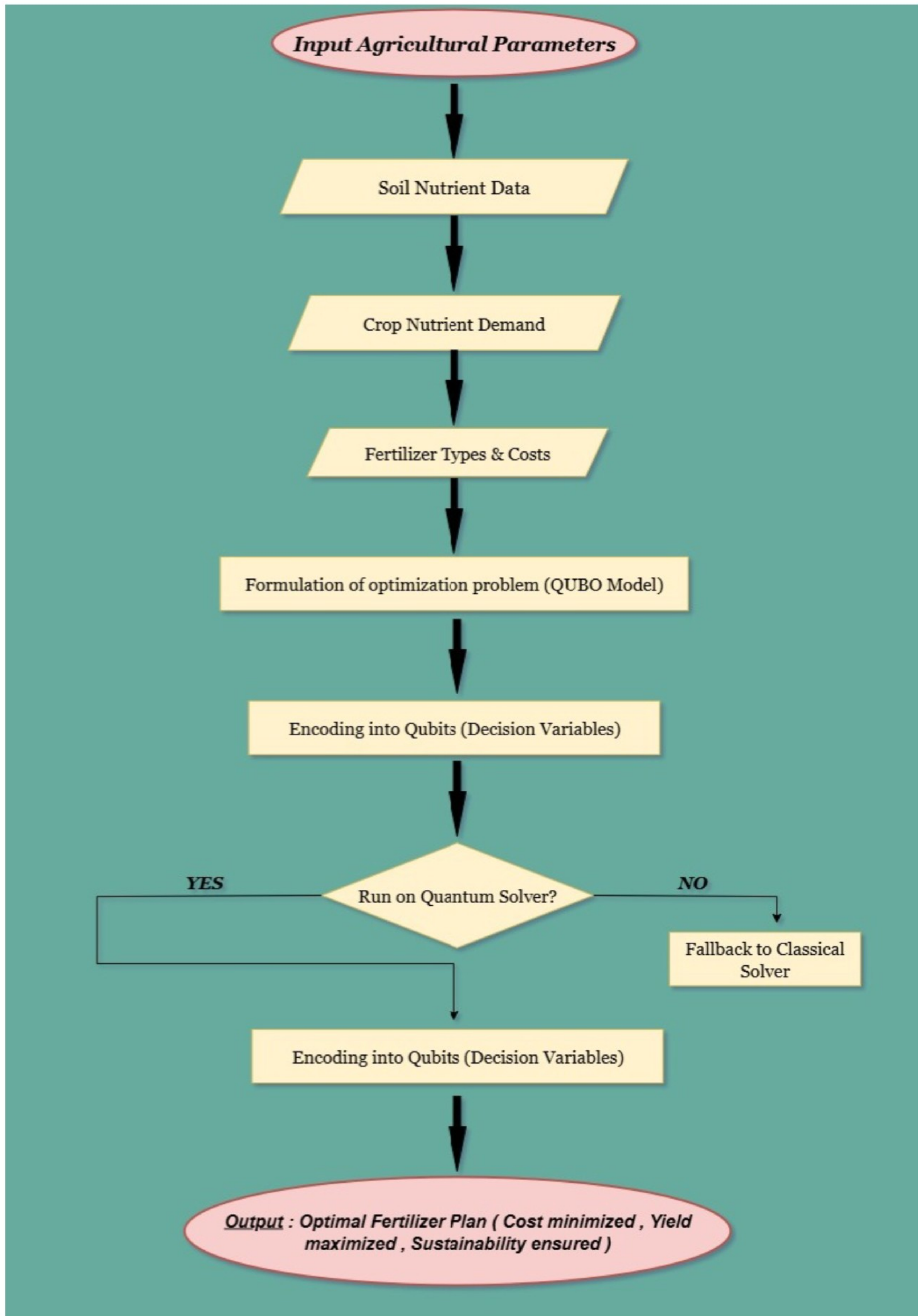


Figure 10: This flowchart illustrates the step-by-step process of leveraging quantum computing for fertilizer optimization, from data collection and problem formulation to simulation and results analysis

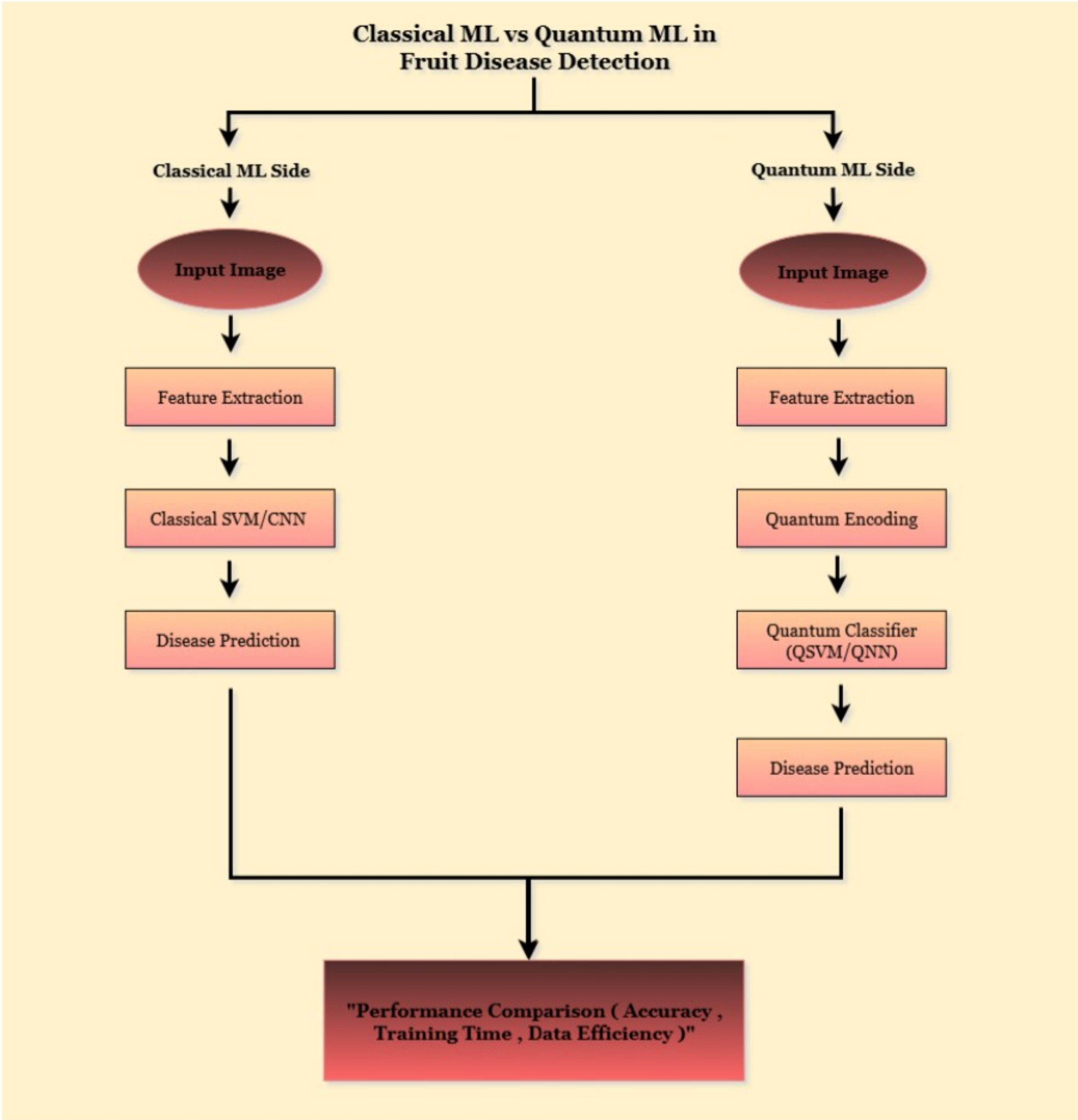


Figure 11: Comparison of Quantum and Classical Machine Learning for Fruit Disease Detection

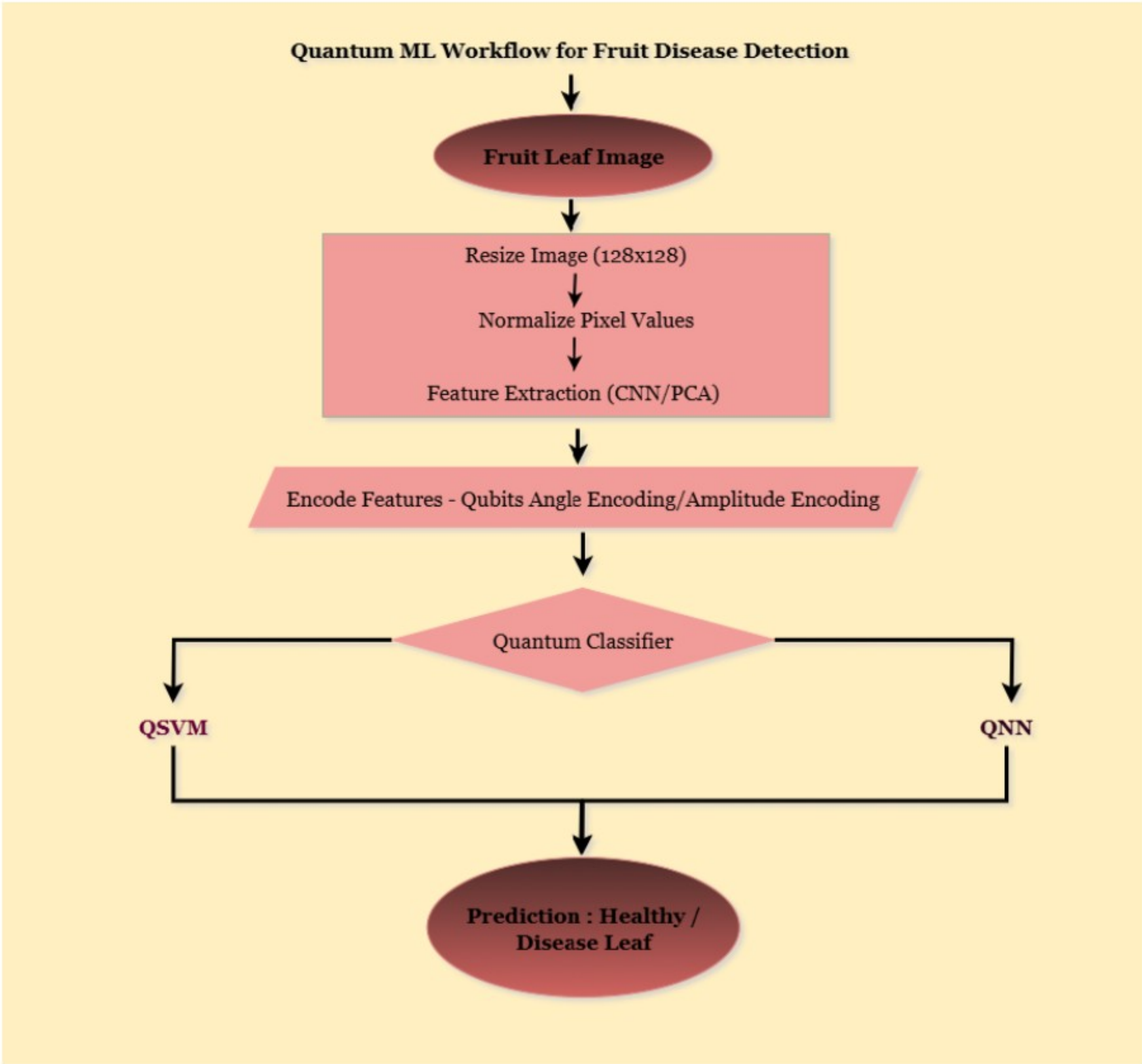


Figure 12: Quantum ML Workflow

10 Conclusion

The integration of quantum computing offers a powerful solution to major agricultural challenges: resource optimization and disease detection.

We demonstrated that framing fertilizer optimization as a scalable QUBO problem, solvable by Quantum Annealers (QA) or QAOA, enables efficient, large-scale decision-making for profit maximization.

For plant disease detection, the proposed Quantum Machine Learning (QML) framework, utilizing angle encoding with QSVMs and QNNs, leverages superposition and entanglement. This approach is poised to overcome classical limitations by achieving higher accuracy and faster training times for real-time diagnostics.

In essence, quantum computing is validated as a foundational tool that can significantly enhance efficiency and sustainability, driving the future of agriculture.

11 Additional Resources

- **Dataset to Train the Model:**

- <https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset>
- https://figshare.com/articles/dataset/PlantVillage_dataset/28234004?file=51801608
- <https://data.mendeley.com/datasets/tywbtsjrjv/1>

- **QNN Resources:**

- https://qiskit-community.github.io/qiskit-machine-learning/tutorials/01_neural_networks.html

- **Understanding the Calculation:**

- <https://arxiv.org/pdf/1811.11538> (Mathematical Calculation)
- https://pennylane.ai/qml/demos/tutorial_QUBO (Understanding QUBO)

- **QSVM:**

- https://www.researchgate.net/publication/393180484_COMPARATIVE_ANALYSIS_OF_CLASSICAL_AND_Q

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